INDEPENDENT CASE STUDY | PUBLIC INFORMATION ONLY

# Union Pacific / Norfolk Southern

## The first transcontinental railroad: is it worth the price, and is the spread mispriced?

M&A and merger-arbitrage assessment of Union Pacific's ~$85B acquisition of Norfolk Southern (1.0 UNP share + $88.82 cash per NSC share)

Valuation date: close of 14-Oct-2026 (UNP $286.43, NSC $323.73) | US$ unless noted

Deal terms, valuation, synergies, accretion, financing, STB review, spread, scenarios and recommendation

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Not prepared for, reviewed by or affiliated with Union Pacific, Norfolk Southern, their advisors, or any employer or client. Not investment advice.

# A sound deal for UNP holders if revenue synergies arrive, and a spread that prices only a 25% chance of closing

View: qualified support for the merger on its merits. Arb: long NSC / short 1.0 UNP, sized so that a break costs 0.5% of NAV.

**25%**

**Market-implied chance of closing**

Versus our 55%. Effective break $315.52 incl. the $2.5B reverse fee; forward to a 31-Oct-27 close.

**1.53x**

**Net synergy NPV / premium**

$19.3B NPV of UP management's $2.0B net synergies vs a $12.6B premium (peer-drift basis).

**+9.4%**

**Year-3 EPS accretion (2030E)**

−1.2% in year 1 and +5.2% in year 2; −1.8% in year 3 with no synergies.

**Full price, real logic.**

Deal value today is $375.25 per NSC share (EV $100B, 17.7x LTM EBITDA): above every rail precedent except CP/KCS (19.5x). Single-line coast-to-coast service is the reason to pay it.

**UNP holders gain $6.9B ($11.56/sh).**

Break-even needs ~$1.1B of net run-rate synergies. The $1.0B of cost synergies alone (NPV $8.8B) falls short, so value depends on revenue synergies.

**Regulation is the swing factor.**

First Class I merger under the 2001 rules, which require applicants to enhance competition. Comments due 18-Nov-2026; final briefs 28-May-2027; decision within 90 days after.

**The arb pays about 3:1.**

Long 1 NSC / short 1 UNP: +$51.24 if it closes, −$17.08 if it breaks. Breakeven 25%. Expected +$20.50/sh (6.3% of notional).

Source: Model (model/UNP_NSC_merger_model.xlsx); merger agreement 28-Jul-2025; UNP 424B3 (Sep-2025); STB Docket FD 36873 (Decision No. 30, 18-Aug-2026); Yahoo Finance closes 14-Oct-2026.

# Transaction overview: 1.0 UNP share plus $88.82 cash, worth $375.25 per NSC share today

Worth $320 at signing (UNP unaffected $231.18), a 25% premium to NSC's 30-day VWAP. UNP is up 24% since, lifting the value.

| Term | Detail |
|---|---|
| Consideration | 1.0 UNP share + $88.82 cash per NSC share. 72% stock at signing; 76% at today's UNP price |
| Value per NSC share | $320.00 at signing (+22.9% vs $260.32 unaffected close, 16-Jul-2025); $375.25 today (+44.1%) |
| Size | Equity $72.1B / EV $85.0B at signing; equity $84.6B / EV $100.1B today |
| Shares / ownership | 225.4M UNP shares; NSC holders own ~27.5% of the combined company (company: ~27%) |
| Termination fees | $2.5B reverse fee paid by UNP on regulatory failure at the End Date or a final adverse order ($11.13/NSC sh); $2.5B paid by NSC on a change of recommendation or alternative deal |
| Conditions | STB approval without a Materially Burdensome Regulatory Condition; CNA (Mexico) approval; votes done 14-Nov-2025 (~99% for). No financing condition; no voting trust |
| End Date | 28-Jan-2028, extended automatically for STB information orders or tolling (471 days away) |
| Governance | Combined company named Union Pacific, HQ in Omaha (Atlanta stays a core location); Jim Vena is CEO |

SOURCES AND USES ($M)

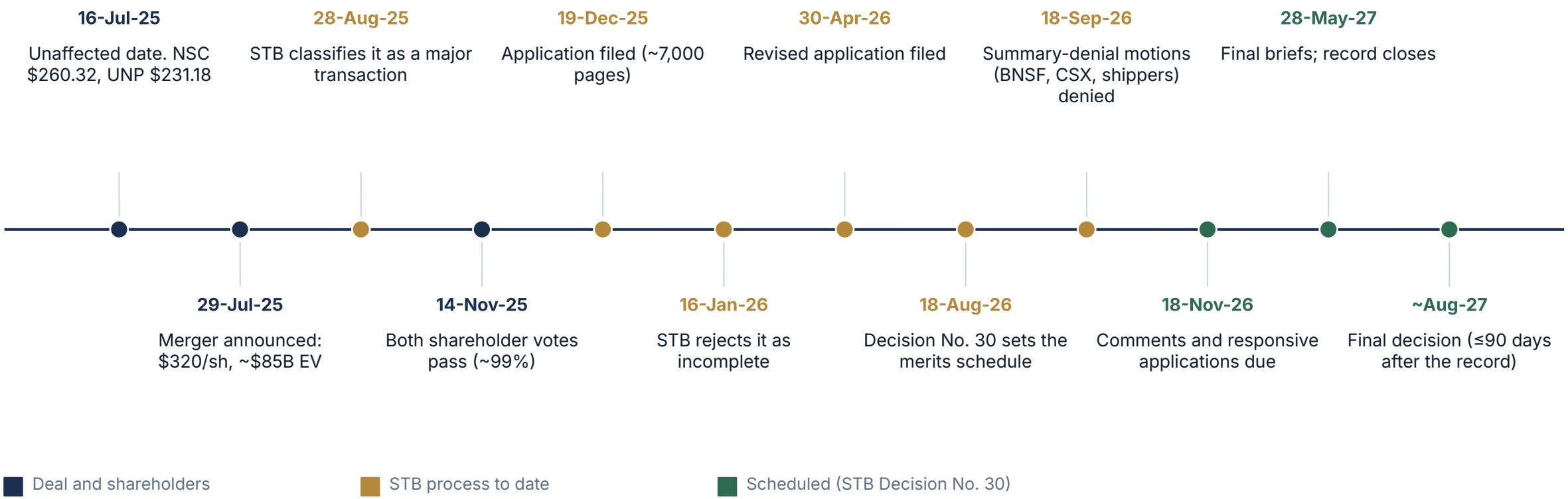
| Uses | $M |
|---|---|
| Cash consideration | 20,020 |
| Financing and other costs | 190 |
| **Total cash uses** | **20,210** |
| **Sources** | |
| UNP cash built before close* | 5,000 |
| New acquisition debt | 15,210 |
| UNP stock to NSC holders | 64,561 |

* Assumption: with buybacks paused, UNP builds ~$5B of cash before closing. The S-4 pro formas assume new debt at 5.0% and $190M of costs; Morgan Stanley used $20.3B of transaction debt.

Source: Agreement and Plan of Merger 28-Jul-2025 (s.2.1, 6.3, 7.1, 7.3); joint press release and investor presentation 29-Jul-2025; 424B3 (pro forma note 6; financing). Model sheets Deal, Financing.

# Fifteen months in: the merits review began 18-Aug-2026 and a decision is due by about August 2027

The first application was rejected as incomplete and the revised one was accepted. Motions for summary denial were denied on 18-Sep-2026.




Implication: the companies first targeted closing by early 2027; UNP's latest 10-Q says 2027. We assume 31-Oct-27 (decision ~Aug-2027, effective ~30 days later), with a delayed case on 30-Apr-28. The End Date (28-Jan-2028) extends automatically for STB information orders or tolling.

Source: STB Docket FD 36873: Decision No. 30 (18-Aug-2026; FR Doc. 2026-17024) and decisions of 28-Aug-2025, 16-Jan-2026, 28-May-2026, 18-Sep-2026 and 6-Oct-2026; UNP/NSC 8-Ks; 424B3. Hearing date not yet set.

# Strategic rationale: one network across 43 states, built to take freight off the highway

End-to-end: UNP runs west of the Mississippi, NSC east. Coast-to-coast freight is now handed off at Chicago, Memphis, St. Louis or New Orleans.

| 2024 | Union Pacific | Norfolk Southern | Combined |
|---|---|---|---|
| Revenue | $24.3B | $12.1B | $36.4B |
| EBITDA | $12.5B | $5.5B | $18.0B |
| Operating ratio | 59.9% | 66.4% | 62.1% |
| Route miles | 32,880 | 19,200 | >50,000 |
| Employees | 32,400 | 19,600 | ~52,000 |
| Footprint | West | East | 43 states, ~100 ports |

### WHAT THE APPLICANTS SAY THE MERGER DELIVERS

- ~2.1M truckloads a year converted to rail and ~$3.5B of annual shipper savings (revised application)
- New single-line routes that remove interchange delays at the Mississippi gateways
- ~1,200 net new union jobs pledged; $2.1B of incremental capital in the application

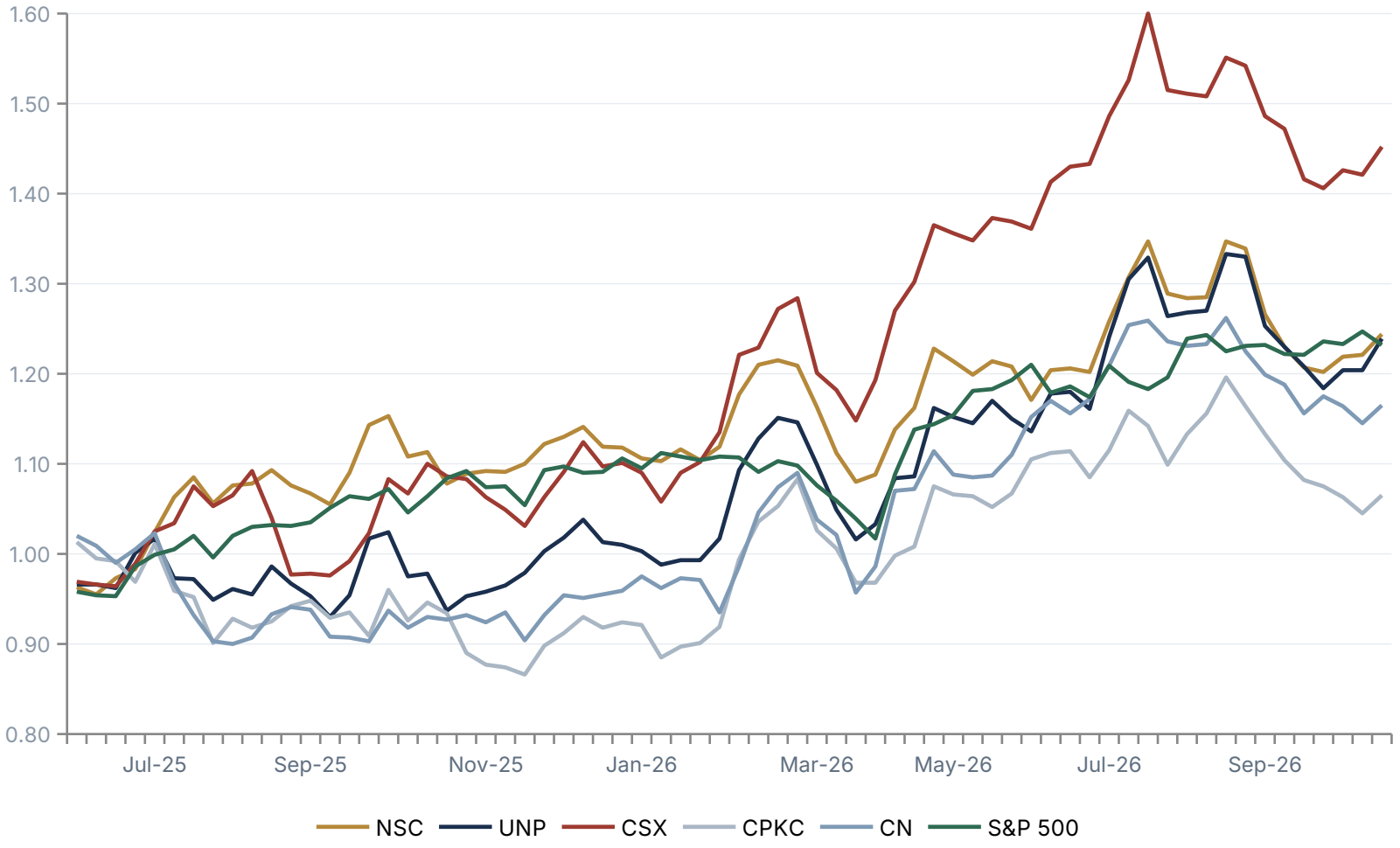
### OUR READ

- Real: this closes the last coast-to-coast gap in US rail. Single-line service competes with long-haul trucking on transit time, and an end-to-end deal has few 2-to-1 overlaps.
- Defensive: once UNP and NSC combine, BNSF and CSX face pressure to respond. The 2001 rules make the STB weigh that ("cumulative impacts and crossover effects").
- Unproven: about 64% of the $2.75B gross synergies are revenue synergies ($1.75B). Rail revenue-synergy claims have a mixed record, and STB conditions can pass part of them to shippers.
- Execution: integrating two networks is a known risk. UP/SP (1997) and the Conrail split (1999) both caused service crises, and the STB weighs transition risk explicitly.

Source: UNP/NSC investor presentation 29-Jul-2025 (2024 figures, pro forma); STB application 19-Dec-2025 and revised application 30-Apr-2026 (truck conversion, shipper savings, jobs, capital). 2024 figures are company-reported, not LTM.

# Since the unaffected date NSC is up 24%, in line with UNP (24%) and well behind CSX (45%)

Prices rebased to 1.00 on 16-Jul-2025. CSX has re-rated on consolidation speculation, which inflates any break value that is drifted with the peers.




CHANGE SINCE 16-JUL-2025

| Stock | Close | Change |
|---|---|---|
| NSC | $323.73 | +24.4% |
| UNP | $286.43 | +23.9% |
| CSX | $48.28 | +45.2% |
| CPKC (US$) | $86.12 | +6.5% |
| CN (US$) | $119.85 | +16.5% |
| S&P 500 | 7,719 | +23.2% |
| Peer basket (eq.-wt.) | | +22.7% |

UNP fell on the announcement, as the market read the price as full, and has since tracked the group. The 10-year Treasury yield rose from 4.46% to 5.19%, which raises discount rates for both companies.

Source: Yahoo Finance weekly closes, 6-Jun-2025 to 14-Oct-2026 (CPKC and CN are NYSE US$ lines). Model sheets Synergies (peer drift) and SpreadHistory.

# Standalone profiles: UNP earns a 50% EBITDA margin; NSC earns 45% and is running 9% below its 2026 plan

LTM to 30-Jun-2026 actuals and the UP management projections in the 424B3 (prepared mid-2025). We haircut NSC plan EBITDA by 6% throughout.

| LTM 30-Jun-2026 ($M) | UNP | NSC |
|---|---|---|
| Revenue | 25,410 | 12,540 |
| Adjusted EBITDA | 12,719 | 5,651 |
| EBITDA margin | 50.1% | 45.1% |
| Operating ratio (NSC adjusted) | 60.0% | 66.2% |
| Adjusted diluted EPS | $12.27 | $12.68 |
| Net debt | 28,713 | 15,547 |
| Net debt / EBITDA | 2.26x | 2.75x |
| EV / LTM EBITDA (market) | 15.6x | 15.6x* |
| P / LTM adjusted EPS | 23.3x | 25.5x* |

* NSC trades on the deal terms, not on its own fundamentals.

UP MANAGEMENT PROJECTIONS ($B; EPS IN $)

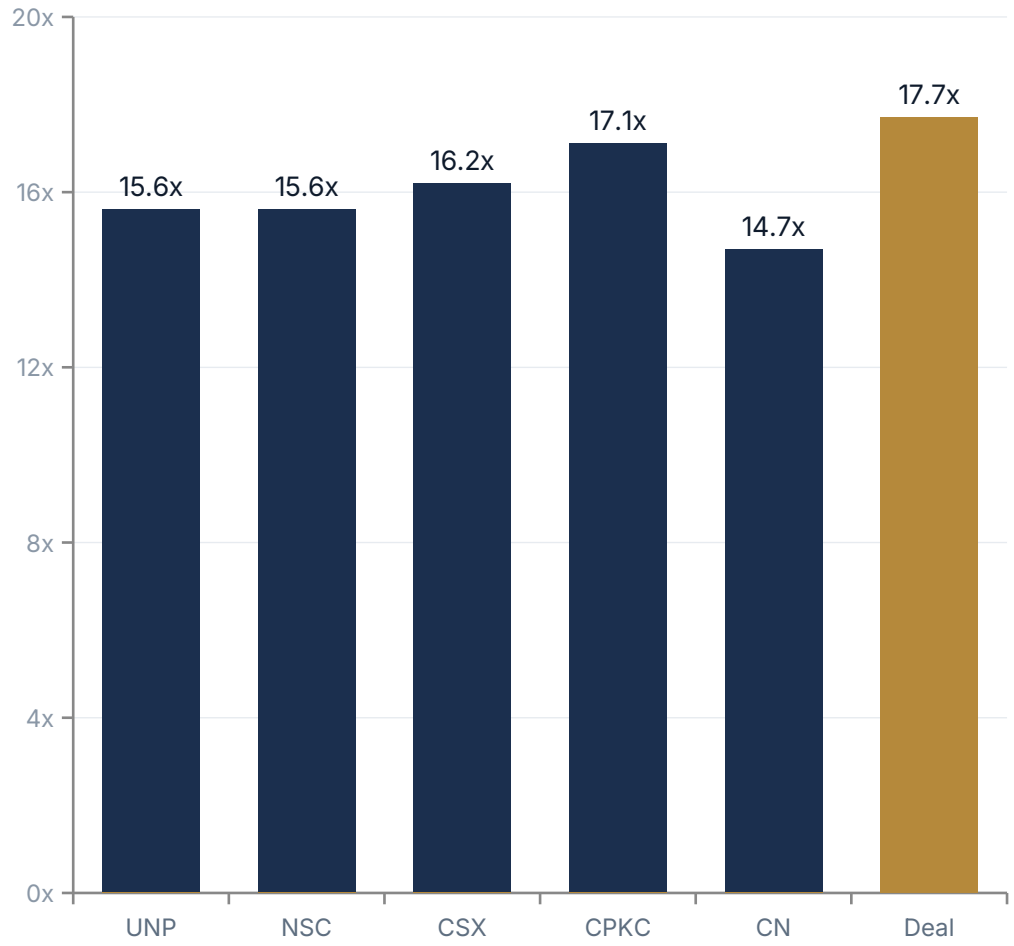
| | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 |
|---|---|---|---|---|---|---|
| UNP revenue | 25.2 | 26.1 | 27.1 | 28.0 | 29.1 | 30.1 |
| UNP EBITDA | 13.1 | 13.7 | 14.2 | 14.8 | 15.4 | 16.0 |
| UNP capex | 3.5 | 3.6 | 3.7 | 3.9 | 4.0 | 4.1 |
| UNP EPS | 12.57 | 13.52 | 14.46 | 15.50 | 16.65 | 17.84 |
| NSC revenue | 12.9 | 13.4 | 14.0 | 14.5 | 15.0 | 15.6 |
| NSC EBITDA | 6.2 | 6.7 | 7.1 | 7.5 | 7.8 | 8.0 |
| NSC capex | 2.2 | 2.2 | 2.2 | 2.3 | 2.3 | 2.4 |
| NSC EPS | 14.05 | 15.72 | 17.33 | 19.11 | 20.86 | 22.47 |

- UNP: revenue grows ~3.6% and EBITDA ~4.1% a year to 2031. LTM EBITDA is 2.9% short of the 2026 plan.
- NSC: the plan lifts EBITDA margin from ~48% to ~51%. That is ambitious against an LTM 45% and a 66% adjusted operating ratio, hence our haircut.

Source: UNP and NSC 10-K 2025, 10-Q Q2-2026 and earnings releases (adjusted EPS, NSC non-GAAP items, UNP adjusted EBITDA); 424B3 “Certain Unaudited Prospective Financial Information” (UP management case for both companies). Model sheets Financials, DCF_U, DCF_N.

# Trading comps: Class I rails trade at 14.7x–17.1x EBITDA; the deal values NSC at 17.7x

NSC has traded at a discount to the group (6.2% assumed). Private BNSF is shown at UNP's multiple as a reference point.




## MULTIPLES AT THE VALUATION DATE

| Company | EV ($M) | EV/EBITDA | P/E | EBITDA margin |
|---|---|---|---|---|
| UNP | 198,874 | 15.6x | 23.3x | 50.1% |
| NSC (deal-affected) | 88,259 | 15.6x | 25.5x | 45.1% |
| CSX | 106,911 | 16.2x | 27.9x | 45.5% |
| CPKC | 93,142 | 17.1x | 28.5x | 50.0% |
| CN (FY2025) | 88,314 | 14.7x | 22.5x | 49.3% |
| BNSF (private; at UNP multiple) | 176,905 | 15.6x | 27.0x | 46.1% |
| Peer median (CSX, CPKC, CN) | | 16.2x | 27.9x | |

## IMPLIED NSC STANDALONE VALUE

| Method | $/NSC share |
|---|---|
| Peer median EV/EBITDA less NSC discount | $310.04 |
| Peer median P/E less discount | $331.93 |
| Range (peer low to high multiple) | $275.80–$359.09 |

Source: EDGAR XBRL company facts (10-K 2025 and 10-Q Q2-2026; CN FY2025 6-K/40-F, no 2026 XBRL); Yahoo Finance closes 14-Oct-2026; USD/CAD 1.4213. EBITDA = operating income + D&A (UNP and NSC: company adjusted). Model sheet Comps.

# DCF: NSC is worth ~$309 and UNP ~$262 standalone, both below market; the deal pays 21% over NSC's DCF

Unlevered DCF of UP management projections (NSC EBITDA haircut 6%), mid-year discounting from 14-Oct-2026, terminal value at 14.5× 2031 EBITDA.

| $M unless noted | UNP | NSC |
|---|---|---|
| WACC | 8.69% | 8.52% |
| PV of FCF, Q4-26 to 2031 | 33,913 | 14,283 |
| PV of terminal value | 150,259 | 71,204 |
| Enterprise value | 184,172 | 85,487 |
| Less net debt (NSC: plus $550M Eastern Ohio) | 28,713 | 16,097 |
| Equity value | 155,459 | 69,390 |
| Value per share | $261.68 | $308.94 |
| vs share price | −8.6% | −4.6% |
| Terminal value share of EV | 82% | 83% |
| Implied perpetual growth | 4.7% | 4.9% |
| Gordon cross-check at 3% growth | $183.07 | $196.15 |

NSC $/SHARE (GREEN: AT OR ABOVE THE $323.73 PRICE)

| WACC \ exit multiple | 13.0x | 14.0x | 14.5x | 15.0x | 16.0x |
|---|---|---|---|---|---|
| −1.0% | $292 | $315 | $326 | $338 | $361 |
| −0.5% | $284 | $306 | $317 | $329 | $351 |
| +0.0% (base) | $276 | $298 | **$309** | $320 | $342 |
| +0.5% | $269 | $290 | $301 | $311 | $333 |
| +1.0% | $261 | $282 | $293 | $303 | $324 |

UNP $/SHARE (GREEN: AT OR ABOVE THE $286.43 PRICE)

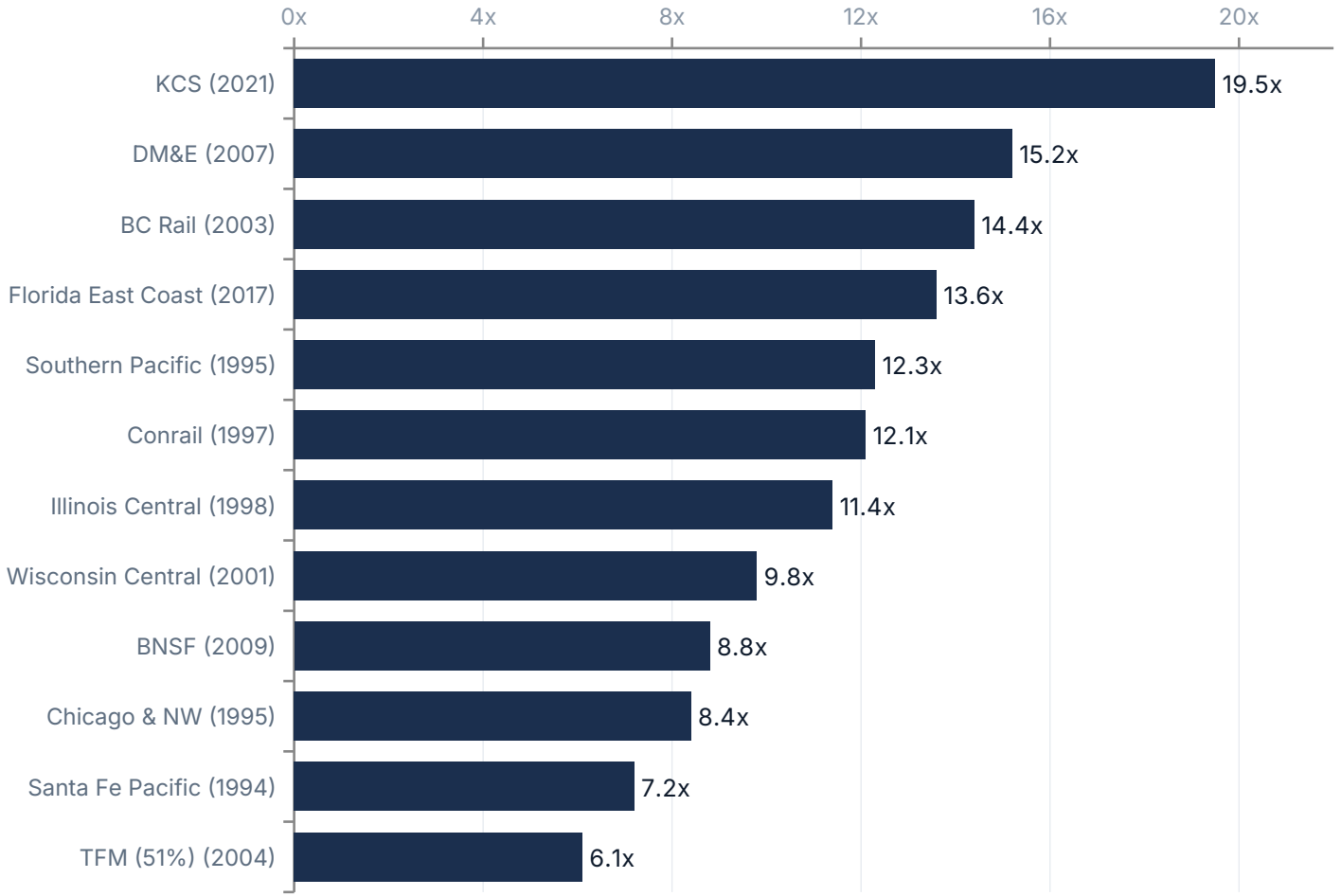
| WACC \ exit multiple | 13.0x | 14.0x | 14.5x | 15.0x | 16.0x |
|---|---|---|---|---|---|
| −1.0% | $248 | $266 | $276 | $285 | $303 |
| −0.5% | $242 | $260 | $268 | $277 | $295 |
| +0.0% (base) | $236 | $253 | **$262** | $270 | $288 |
| +0.5% | $230 | $247 | $255 | $264 | $281 |
| +1.0% | $224 | $240 | $249 | $257 | $274 |

UNP's price implies an exit multiple of ~15.9x, above our 14.5x. Rail valuations depend on terminal multiples holding near today's levels.

Source: 424B3 projections. WACC: 10-yr Treasury 5.19% (^TNX 14-Oct-2026); beta 1.0 and ERP 4.25% (assumptions); Kd 5.9%; tax 23.5%. Advisor ranges ex-synergies: MS $274–342, WF $270–350 per NSC share. Model sheets DCF_U, DCF_N.

# Precedents: at 17.7x LTM EBITDA today, the deal sits above every rail deal except CP/KCS (19.5x)

At signing the price was 14.0x NTM EBITDA. The 12-deal median is 11.75x and the 2001+ mean 12.5x. Scarcity value has risen as the Class I group shrank to six.




## NSC VALUE AT PRECEDENT MULTIPLES

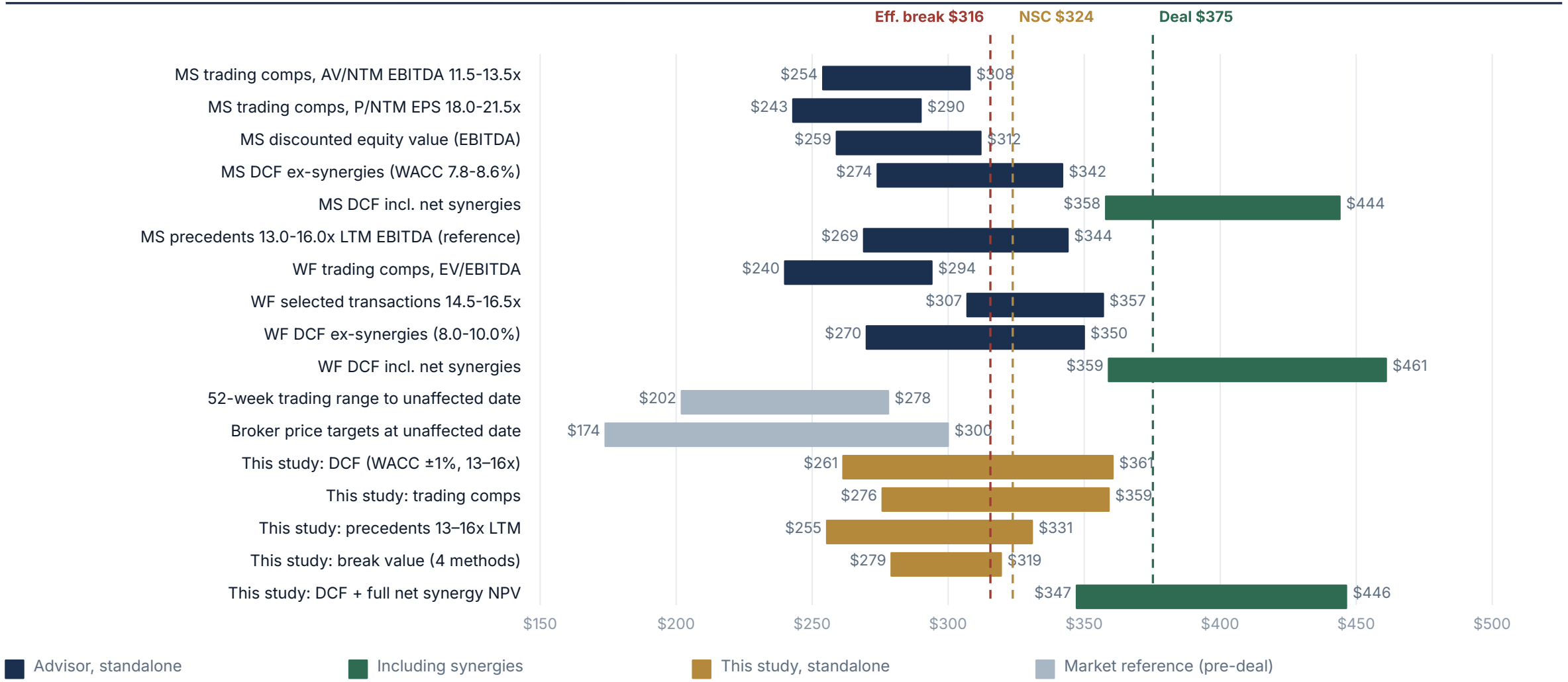
| Multiple of LTM EBITDA | $/NSC sh |
|---|---|
| Median 11.75x | $223.96 |
| 13.0x (MS low) | $255.40 |
| 16.0x (MS high) | $330.88 |
| 19.5x (CP/KCS) | $418.94 |
| Deal value today (17.7x) | $375.25 |

- The 1994–2004 deals were struck when rail margins were far lower; today's operating ratios are ~60–66%.
- CP/KCS is the closest comparator: a scarce end-to-end network, reviewed under the old rules.
- BNSF (2009) was bought near the cycle trough, at 8.8x.

Source: 424B3, Opinion of Wells Fargo Securities (selected transactions, TEV / LTM EBITDA); Morgan Stanley referenced 13.0–16.0x. NSC LTM adj. EBITDA $5,651M (30-Jun-2026). Model sheet Precedents.

# Football field: standalone methods centre on $270–340; only values that include synergies reach the $375 deal value

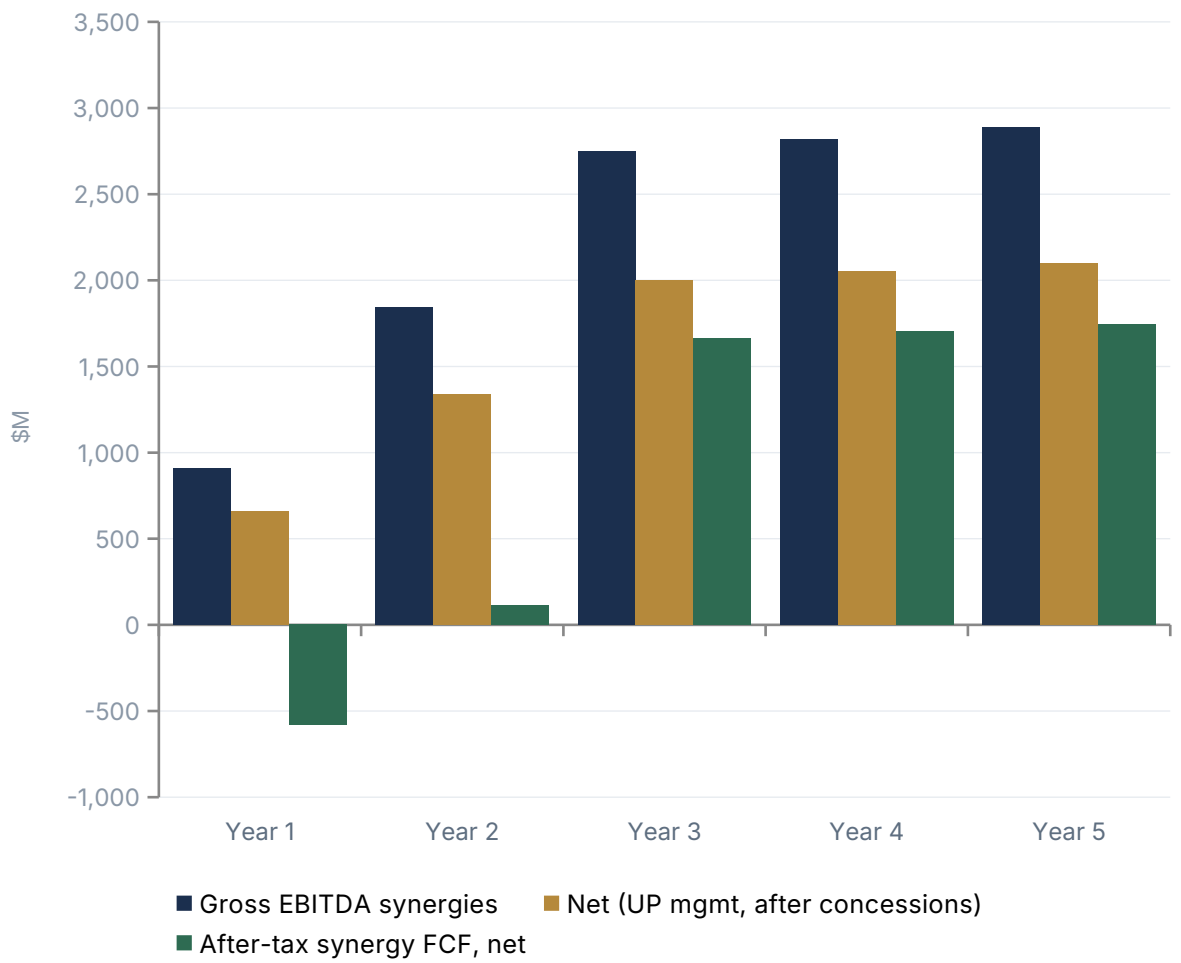
$ per NSC share. Advisor ranges are from mid-2025 fairness opinions; ours are at the valuation date. Dashed lines: effective break, NSC close and deal value.




Source: 424B3: Opinions of Morgan Stanley and Wells Fargo (advisor ranges, 52-week range and broker targets to 16-Jul-2025). This study: model sheets DCF_SUM, Comps, Precedents, Arb, Football.

# Synergies: $2.75B gross and $2.0B net of concessions, worth $19.3B today against a $12.6B premium

Company target: run-rate by the end of year 3. We phase 33/67/100%, grow 2.5% a year after, and discount at 9% (WACC plus execution risk).




## NPV OF SYNERGIES VS PREMIUM ($M)

| Measure | $M | x premium |
|---|---|---|
| Gross synergy NPV ($2.75B run-rate) | 26,521 | 2.11x |
| Net synergy NPV ($2.0B, base) | 19,277 | 1.53x |
| Cost synergies only ($1.0B) | 8,781 | 0.70x |
| Premium today (vs peer-drifted $260.32) | 12,572 | 1.00x |
| Premium at signing ($320 vs $260.32) | 13,452 | |
| Company claim (“>$30B value creation”) | 30,000 | |

## BREAK-EVEN SYNERGIES

- Net run-rate needed to cover today’s premium: $1.36B, leaving $1.39B of the gross figure for concessions and slippage.
- Revenue synergies are 64% of the gross. Without them the deal does not cover its premium.

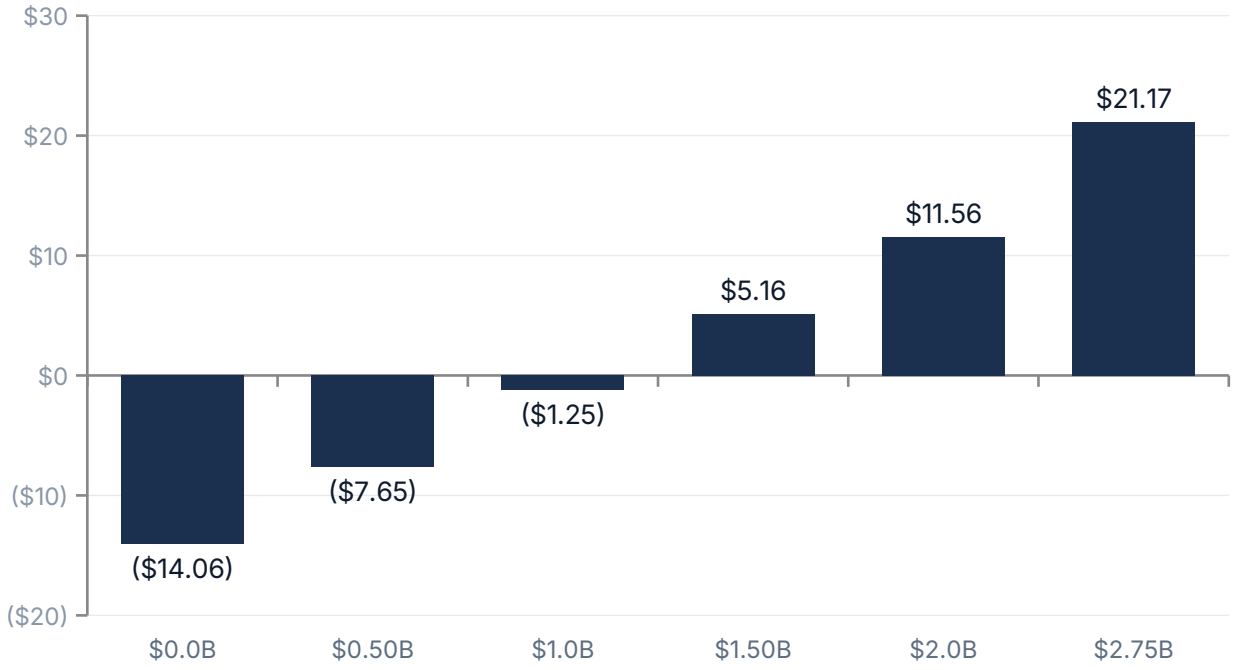
Source: 424B3 and investor presentation 29-Jul-2025 ($1.75B revenue / $1.0B cost synergies; $170M opex and $2.0B capex to achieve; UP management $2.0B net); STB application ($133M capital synergies). Phase-in, growth and discount rate are assumptions. Model sheet Synergies.

# For UNP holders: +$6.9B (+$11.56 a share) on $2.0B of net synergies; break-even at $1.1B

Has / gets: UNP holders give up 27.5% of the combined equity and pay the cash; in return they get NSC and the synergies. Standalone values are our DCFs.

| Has / gets ($M) | Value |
|---|---|
| UNP standalone equity (DCF) | 155,459 |
| + NSC standalone equity (DCF) | 69,390 |
| + Net synergy NPV | 19,277 |
| – Cash paid and costs | (20,210) |
| **= Pro forma equity** | **223,916** |
| x UNP holders' share | 72.5% |
| **= UNP holders get** | **162,327** |
| **Value created (gets – has)** | **6,869** |
| As % of UNP standalone value | +4.4% |
| NSC holders receive per share (DCF basis) | $362.06 |

VALUE CREATED PER UNP SHARE, BY NET RUN-RATE SYNERGIES

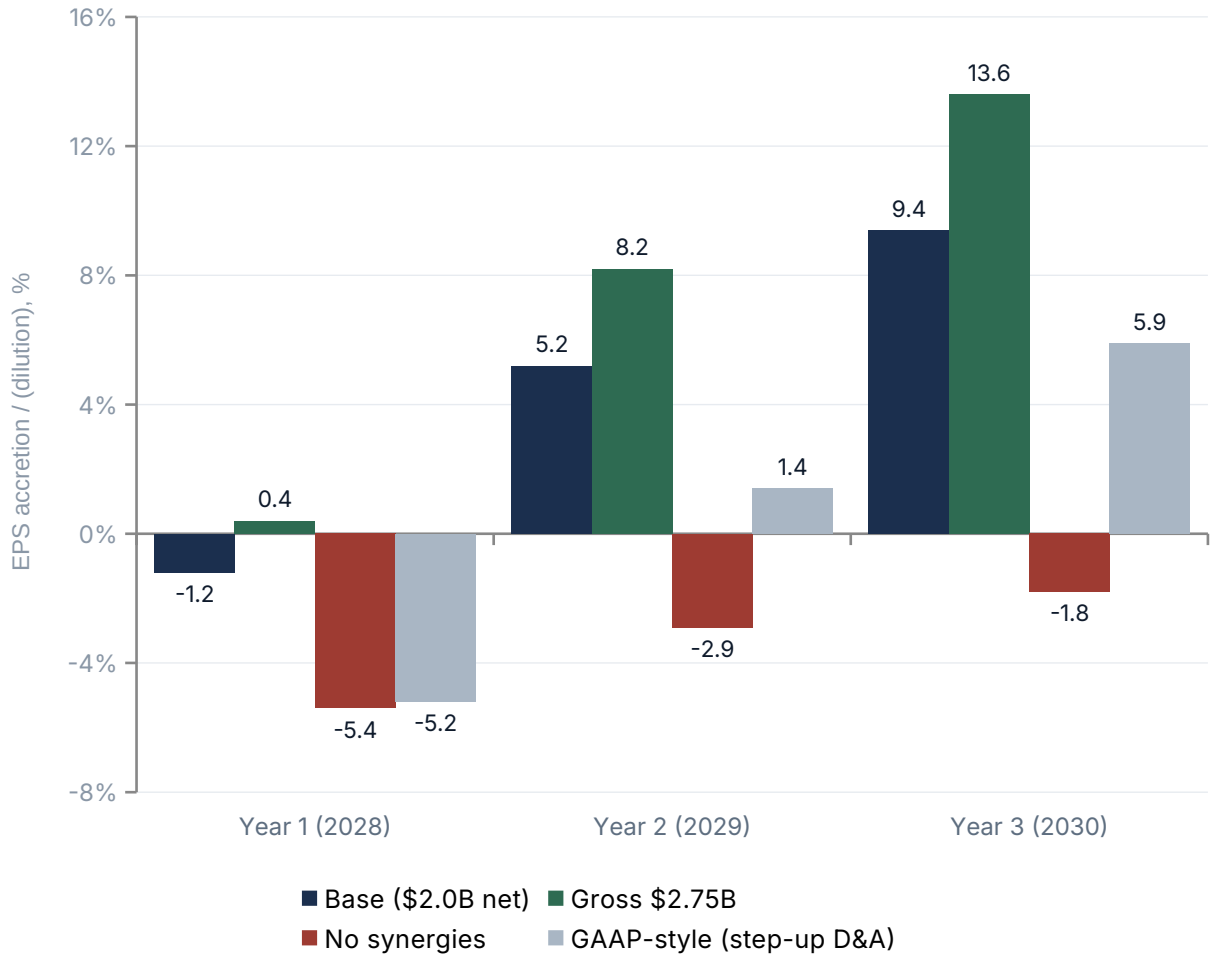



- Each $100M of net run-rate synergies is worth ~$1.05B of NPV, or ~$1.28 per UNP share.
- With no synergies UNP holders lose $8.4B: NSC is bought 21% above its DCF value.
- Verdict: positive, but it depends on execution and on the concessions the STB imposes.

Source: Model sheet Synergies (has/gets block); DCF values from sheets DCF_U and DCF_N. Cash = $88.82 × 225.4M shares + $190M of costs. NSC holders' receipt = $88.82 + pro forma value per share.

# EPS: −1.2% in year 1, +5.2% in year 2 and +9.4% in year 3, in line with company guidance

Adjusted EPS (ex purchase-accounting amortisation); close 31-Oct-27. Guidance: accretive early in year 2, high single digits in year 3+.




**PRO FORMA EPS BUILD ($M)**

| | 2028 | 2029 | 2030 |
|---|---|---|---|
| UNP standalone net income | 8,589 | 9,207 | 9,890 |
| NSC net income (haircut) | 3,573 | 3,956 | 4,335 |
| Net synergies after tax | 505 | 1,025 | 1,530 |
| Cost to achieve after tax | -130 | 0 | 0 |
| Interest on new debt after tax | -669 | -669 | -669 |
| Lost interest on cash used | -156 | -156 | -156 |
| **Pro forma net income** | **11,712** | **13,362** | **14,930** |
| Pro forma shares (M) | 819.4 | 819.4 | 819.4 |
| **Pro forma EPS** | **$14.29** | **$16.31** | **$18.22** |
| UNP standalone EPS | $14.46 | $15.50 | $16.65 |
| **Accretion / (dilution)** | **−1.2%** | **+5.2%** | **+9.4%** |
| Pre-tax synergies for EPS-neutral | 838 | 475 | 317 |

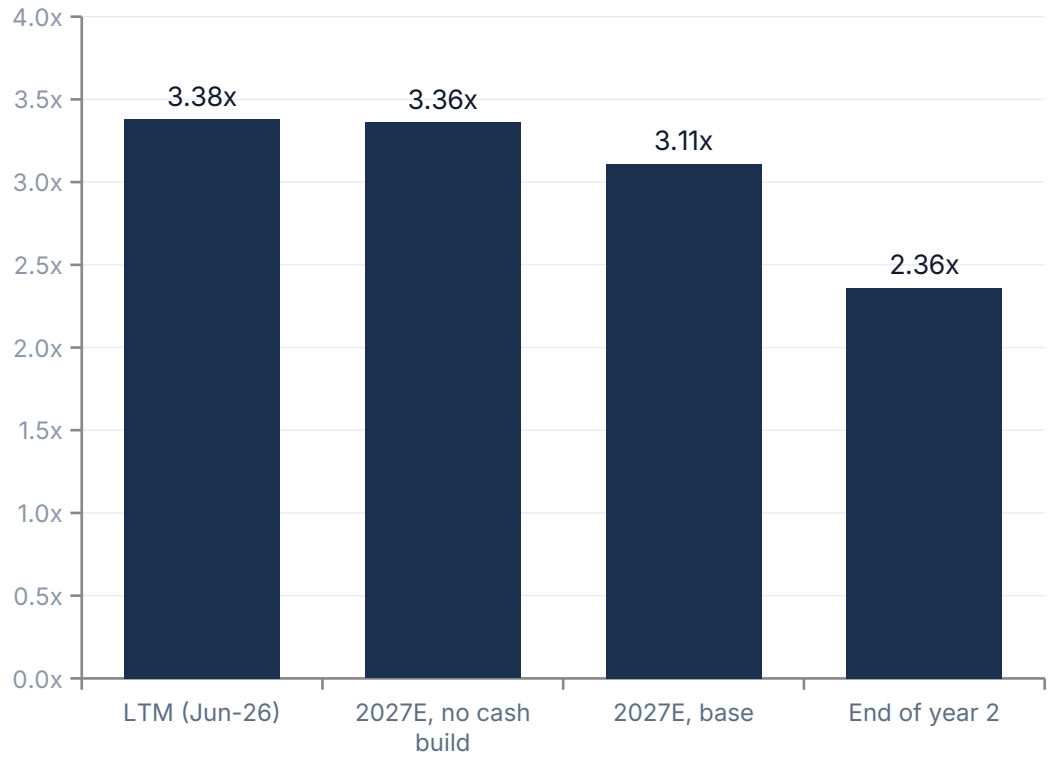
GAAP-style line adds an illustrative $25B PP&E step-up depreciated over 40 years (assumption; the S-4 allocation is preliminary).

Source: 424B3 UP management EPS projections (UNP and NSC); investor presentation 29-Jul-2025 (guidance). New debt $15.2B at 5.75% (assumption); NSC earnings haircut 6%; 225.4M new shares; share counts held flat. Model sheet ProForma.

# Financing: ~$15B of new debt takes leverage to ~3.1x at close, falling to ~2.4x by year 2 before buybacks

UNP has committed to an "A" rating, paused buybacks until year 2 and kept the dividend. We assume a ~$5B pre-close cash build and no buybacks in years 1–2.

## PRO FORMA GROSS DEBT / EBITDA




Guidance: ~3.3x at close and ~2.8x by year 2 (with buybacks resumed). UNP alone today: 2.26x net debt / EBITDA.

## PRO FORMA CREDIT PROFILE ($M)

| Measure | Value |
|---|---|
| UNP + NSC debt (30-Jun-26) | 46,943 |
| + New acquisition debt | 15,210 |
| **Pro forma gross debt** | **62,153** |
| Pro forma 2027E EBITDA (no synergies) | 19,998 |
| FCF after dividends, year 1 / year 2 | 4,423 / 5,919 |
| EBITDA / interest (2027E) | 6.8x |
| After-tax cost of new debt (a year) | 669 |

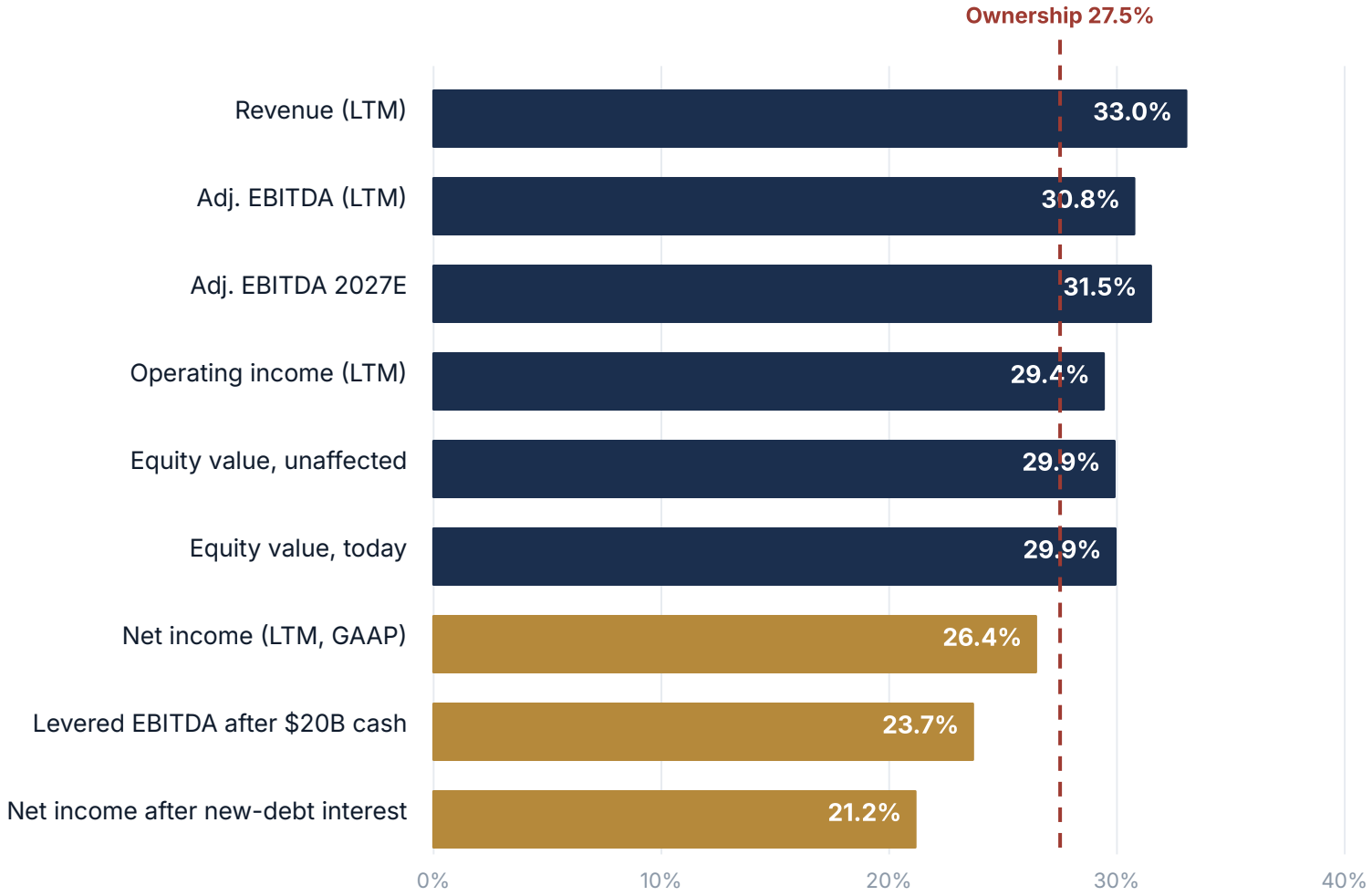
## RATINGS

- UNP: A3 / A- / A- (Moody's / S&P / Fitch, Feb-2025 FWP). Fitch put UNP on Rating Watch Positive on 29-Jul-2025 and kept it there on 15-Jul-2026.
- NSC: S&P outlook positive. Moody's Baa1 on review for upgrade (secondary source; not verified).
- At ~3.1–3.4x, holding the "A" needs fast deleveraging; the timing of the year-2 buyback restart is the swing factor.

Source: Investor presentation 29-Jul-2025 (~3.3x at close, ~2.8x by year 2, buybacks resume in year 2 and reach $10B+ a year by year 3); 10-Q Q2-2026 balance sheets; FWP Feb-2025 (UNP A3/A-/A-); Fitch rating actions 29-Jul-2025 and 15-Jul-2026. Model sheet Financing.

# Contribution: NSC brings ~31% of EBITDA for 27.5% of the shares, plus $20B of cash

Unlevered, NSC contributes more than its ownership. Charge NSC holders for the cash they receive and the levered contribution falls to 24%, below 27.5%.




### READING THE ANALYSIS

- NSC holders get 27.5% of the shares plus $20.0B of cash. Together that is worth more than the 29–31% of EBITDA they contribute: the gap is the control premium.
- NSC's net income share is lower (26%, or 21% after new-debt interest) because it carries more D&A and debt per $ of EBITDA.
- The exchange ratio is fixed, so NSC holders share in UNP's price moves, up or down, until closing.

Source: LTM to 30-Jun-2026 from company filings; 2027E from 424B3 projections (NSC haircut 6%); prices 16-Jul-2025 and 14-Oct-2026. Levered EBITDA contribution at UNP's EV/EBITDA, less each side's net debt and the cash paid. Model sheet Contribution.

# The STB test: under the 2001 rules applicants must show the merger enhances competition, not just preserves it

This is the first combination of two Class I railroads reviewed under the 2001 rules. CPKC (2023) was decided under the earlier rules through a waiver for KCS.

## WHAT THE BOARD MUST FIND

| Element | Standard |
|---|---|
| Public interest (11324(c)) | Approve if consistent with the public interest; broad power to impose conditions |
| Statutory factors (11324(b)) | Adequacy of service; effect of including other carriers; fixed charges; employees; rail competition |
| 2001 policy (1180.1(c)) | "Substantial and demonstrable" public benefits must outweigh anticompetitive effects and service risk |
| Enhanced competition (1180.1(d)) | Applicants "should propose conditions that would not simply preserve but also enhance competition" |
| Crossover effects (1180.1(i)) | Must address how further Class I mergers would affect the industry and the conditions |
| Service and safety (1180.1(f), (h)) | Service Assurance Plan with benchmarks; Safety Integration Plan with the FRA |
| Oversight (1180.1(g)) | At least 5 years of formal oversight; the Board can add conditions |
| Timing (11325(b)) | Evidence closes within 1 year of the notice; final decision within 90 days after that |

## LIKELY CONDITIONS, AND WHAT THE CONTRACT SAYS

- Expect: commitments to keep the Chicago, Memphis, St. Louis and New Orleans gateways open on reasonable terms; trackage rights where shippers lose an option; service benchmarks; extended oversight (CPKC got 7 years); New York Dock labour protection; environmental mitigation.
- Opposition: BNSF, CSX and a shipper coalition (ACC, AFPM, ACD, NITL, TFI) sought summary denial. The STB refused on 18-Sep-2026, calling it a transaction of first impression that needs a full record.
- Contract: UNP must use reasonable best efforts, including divestitures, but need not accept a Materially Burdensome Regulatory Condition. If the End Date passes without STB approval, or a final order blocks the deal, UNP pays NSC $2.5B.
- Our base case: approval with significant but tolerable conditions that use up part of the $0.75B gap between gross and net synergies.

Source: 49 U.S.C. 11321–11328 (esp. 11324(b)–(c), 11325(b)); 49 CFR 1180.1 (Major Rail Consolidation Procedures, EP 582 (Sub-No. 1), 66 FR 32583, 15-Jun-2001); STB decision FD 36500 (15-Mar-2023); STB decision of 18-Sep-2026 denying summary denial.

# Timeline and precedent: CPKC took ~17 months from application to decision under the old rules; UNP/NSC is on a similar path

Decision No. 30 sets the merits schedule. The hearing date is still to be set, and the Board must decide within 90 days of the final briefs.

## PROCEDURAL SCHEDULE (DECISION NO. 30)

| Step | Date |
|---|---|
| Application accepted (held in abeyance) | 28-May-2026 |
| Decision No. 30: schedule adopted | 18-Aug-2026 |
| Notices of intent to participate (extended) | 30-Sep-2026 |
| Comments, responsive applications | 18-Nov-2026 |
| DOJ and USDOT comments | 3-Dec-2026 |
| Responses and rebuttal | 16-Feb-2027 |
| Rebuttal on responsive applications | 29-Mar-2027 |
| Oral argument / hearing | To be set |
| Final briefs | 28-May-2027 |
| Final decision (no later than) | ~26-Aug-2027 |
| Assumed close (this study) | 31-Oct-27 |

## CPKC (2023) VS UNP/NSC

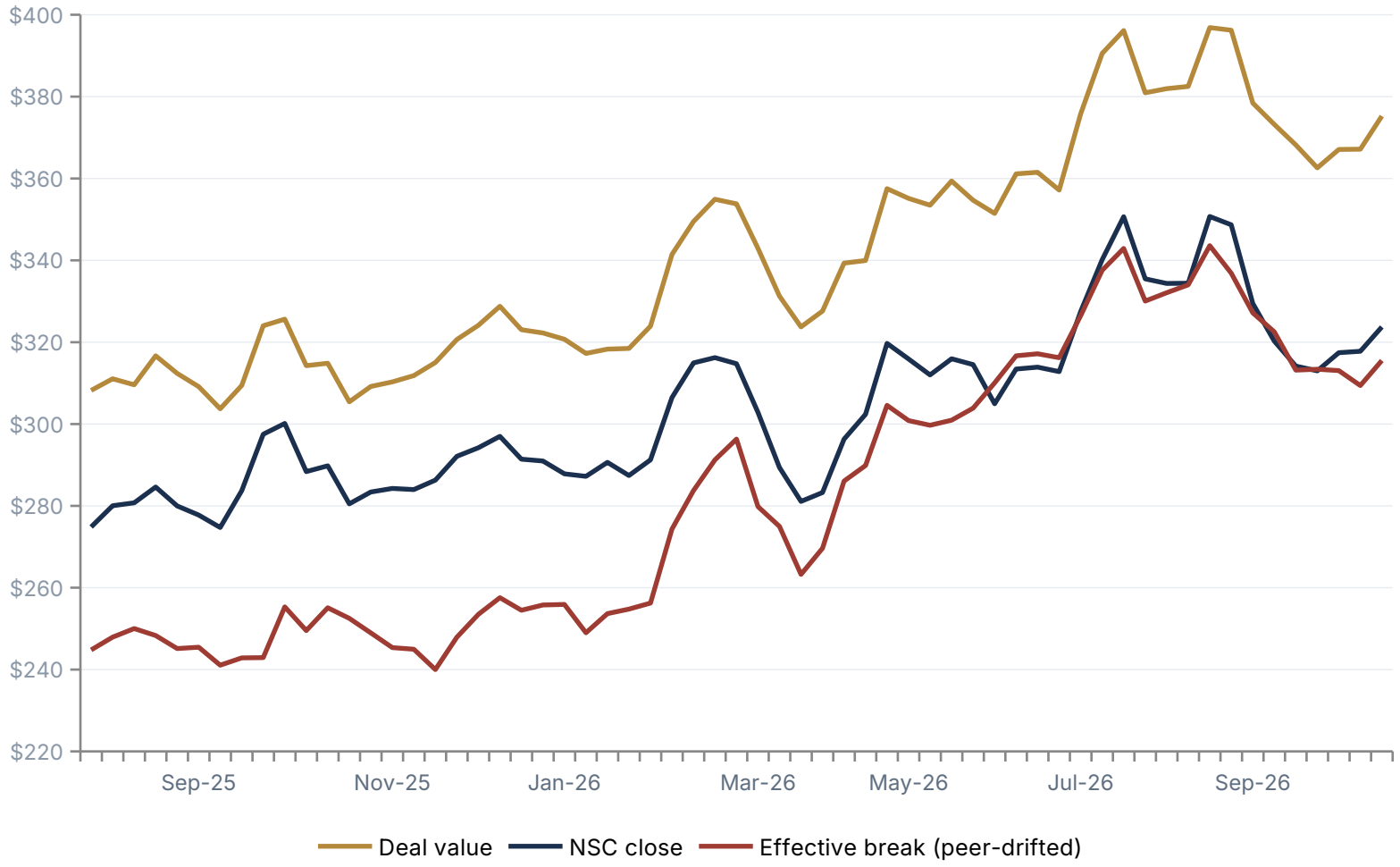
| | CP / KCS | UNP / NSC |
|---|---|---|
| Rules applied | Pre-2001 (KCS waiver) | 2001 rules (enhance competition) |
| Type | End-to-end, 3 countries | End-to-end, US |
| Size (EV at signing) | ~$31B | ~$85B |
| Voting trust | Yes (Dec-2021) | No |
| Application to decision | Oct-2021 to Mar-2023 (~17 mo.) | Apr-2026 to ~Aug-2027 (~16 mo.) |
| Opposition | CN (rival bidder), some shippers, Metra | BNSF, CSX, shipper coalition |
| Outcome | Approved with conditions; 7-yr oversight | Pending |
| Class I railroads after | 6 | 5 |

Read-across: CPKC shows the Board will approve an end-to-end deal with heavy conditions and long oversight. But CPKC faced a lower bar, used a voting trust to protect its sellers, and left six Class I railroads; this deal would leave five.

Source: STB Decision No. 30 (FD 36873, 18-Aug-2026; FR Doc. 2026-17024); STB decisions of 30-Sep-2026 (deadline extension) and 6-Oct-2026; STB FD 36500 decision 15-Mar-2023, effective 14-Apr-2023 (CPKC dates per that decision).

# The spread: $51.52 (15.9%) gross, 15% annualised to a 31-Oct-27 close

NSC weekly closes vs the deal value (UNP x 1.0 + $88.82) and the peer-drifted effective break. The spread widened after the Jan-2026 rejection.




## SPREAD AT THE VALUATION DATE

| Measure | Value |
|---|---|
| Deal value | $375.25 |
| NSC close | $323.73 |
| Gross spread | $51.52 / 15.9% |
| Forward spread (carry, dividends) | $49.65 |
| Annualised (simple) | 15.2% |
| Over T-bill (4.08%) | 11.1% |
| Days to assumed close | 382 |

- Narrowest weekly spread: 8.5% (3-Oct-25).
- Widest: 17.2% (18-Sep-26).
- The stock leg is hedged by shorting 1.0 UNP. The $88.82 cash leg is fixed, so a falling UNP widens the % spread.

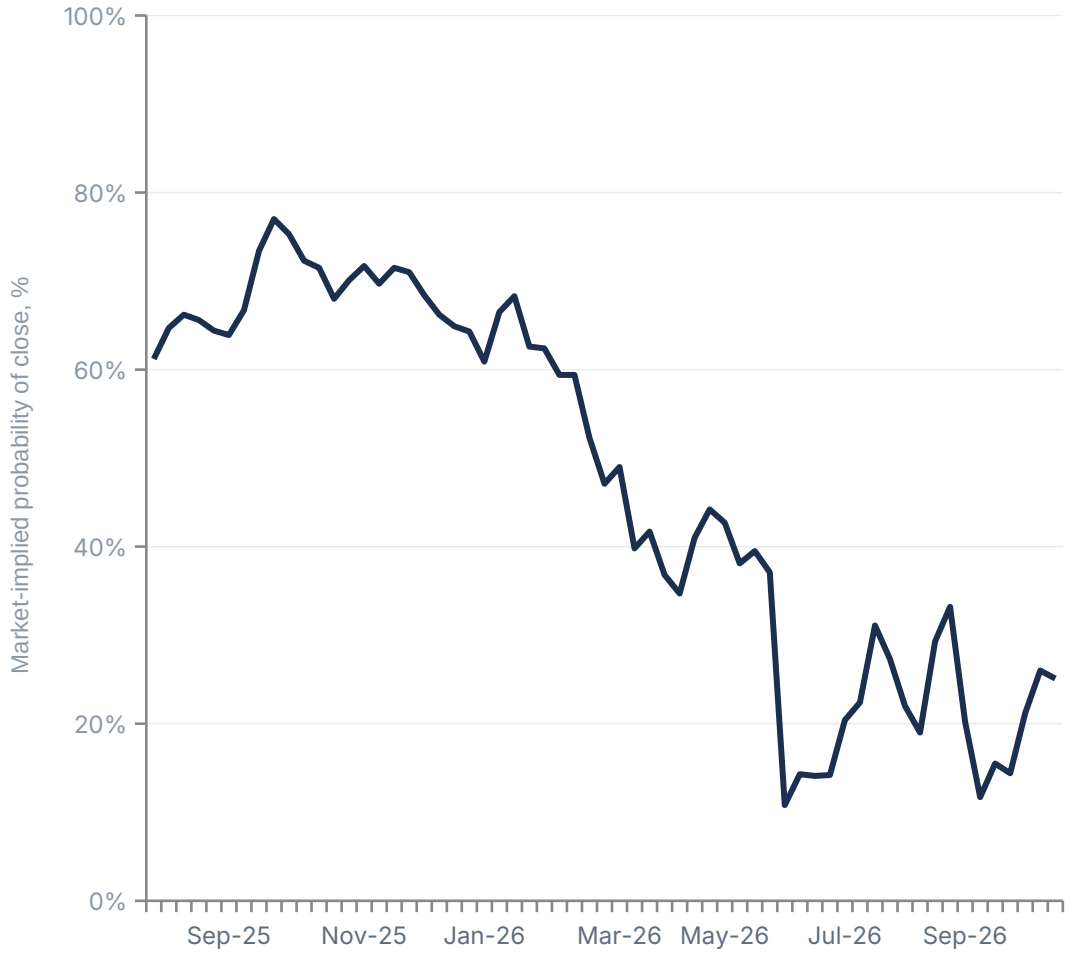
Source: Yahoo Finance weekly (Friday) closes 1-Aug-2025 to 14-Oct-2026; model sheet SpreadHistory. Effective break = $315.52 at the valuation date (incl. the $11.13 reverse fee) x peer index.

# The market implies a 25% chance of closing, down from ~60–75% in late 2025

Implied P = (NSC forward − break) / (deal value forward − break), with the $2.5B reverse fee added to the break. The answer is very sensitive to the break value.

| Break-value method | $/NSC sh | Implied P |
|---|---|---|
| M1 NSC unaffected x peer drift (+22.7%) | $319.48 | 3% |
| M2 Peer median EV/EBITDA less NSC discount | $310.04 | 18% |
| M3 NSC standalone DCF | $308.94 | 20% |
| M4 Unaffected NSC/UNP multiple ratio (0.892) x UNP today | $279.11 | 46% |
| **Base break (average)** | **$304.39** | |
| + Reverse termination fee | $11.13 | |
| **Effective break** | **$315.52** | **25%** |
| Spot basis (no carry) | | 14% |

Our read: M1 is inflated by CSX's takeover-speculation rally, and M4 is the cleanest relative measure. A lower break would raise the implied probability.




Source: Model sheets Arb and SpreadHistory. Forward basis: carry at the 13-week T-bill (4.08%) and four quarterly dividends each side to 31-Oct-27. The history uses that fixed close date, which overstates the horizon for 2025 dates (closing was then expected in early 2027).

# Scenarios: we put the chance of closing at 55%; the hedged trade then earns +$20.50 per NSC share (6.3% of notional)

Four outcomes. The probabilities are our judgement (assumptions). Payoffs per NSC share include dividends to resolution; PV at the T-bill rate.

| | Outcome | Prob. | Payoff | PV | Return |
|---|---|---|---|---|---|
| A | Approved with tolerable conditions; closes 31-Oct-27 | 47% | $380.65 | $365.03 | +17.6% |
| B | Approved after delay; closes 30-Apr-28 | 8% | $383.35 | $360.35 | +18.4% |
| C | Approved with a burdensome condition; UNP walks and pays $2.5B | 5% | $320.92 | $307.75 | −0.9% |
| D | STB denies (or deal withdrawn); $2.5B paid | 40% | $320.92 | $308.80 | −0.9% |
| | **Probability-weighted (unhedged long NSC)** | 100% | $353.99 | $339.30 | +9.3% |

Unhedged long NSC: expected excess return over T-bills +4.8% (PV). Hedged: a $51.24 gain if it closes and a $17.08 loss if it breaks; breakeven P(close) 25%.

Why 55%: major end-to-end and parallel combinations have been approved with conditions (UP/SP 1996, Conrail 1998, CPKC 2023); the Board refused summary denial; the parties have strong incentives to offer remedies. Against: the 2001 bar, organised Class I and shipper opposition, and the rejected first application.

HEDGED EXPECTED P&L, % OF NOTIONAL

| P(close) \ break shift | −$40 | −$20 | +$0 | +$10 |
|---|---|---|---|---|
| 30% | −7.6% | −3.3% | +1.1% | +3.2% |
| 40% | −4.2% | −0.5% | +3.2% | +5.0% |
| 50% | −0.9% | +2.2% | +5.3% | +6.8% |
| 55% (base) | +0.8% | +3.6% | **+6.3%** | +7.7% |
| 60% | +2.4% | +4.9% | +7.4% | +8.6% |
| 70% | +5.8% | +7.6% | +9.5% | +10.4% |

Columns shift the effective break ($/NSC share). The trade loses money if the chance of closing is below ~25% at our base break, or below ~42% if the break is $20 lower.

Source: Model sheet Scenarios. Hedged P&L = long 1 NSC, short 1 UNP. Close: spread + dividend difference. Break: effective break − NSC price − a 3% UNP rise (assumption) + dividend difference.

# Risks and mitigants: regulation dominates for the arb; synergies and leverage dominate for UNP holders

Ranked by expected damage. Impacts per NSC share for the hedged arb, and per UNP share for the deal-merit view.

| Risk | Likelihood | Impact | Mitigant / rule |
|---|---|---|---|
| STB denies, or conditions trigger a walk (MBRC) | **Medium-high** | Hedged arb: −$17.08/sh (−5.3% of notional) | Size to a 0.5% NAV loss (~9% of NAV long notional). The $2.5B fee ($11.13/sh) cushions a regulatory break. |
| Break value lower than modelled (sector de-rates, CSX premium fades) | Medium | Each $10 lower costs ~1.4% of expected return | M4 (relative multiple) gives $279; at that break the trade still earns + $9.12/sh. |
| Delay beyond Aug-2027 (hearing, remand, appeal) | Medium | Lower annualised return; carry on the short | The End Date auto-extends for STB delays; scenario B (8%) prices a six-month slip. Carry is largely matched long and short. |
| Revenue synergies fall short; concessions larger | Medium | UNP holders: value turns negative below ~$1.1B net | Cost synergies ($1.0B) are more controllable. Track STB conditions against the $0.75B gross-to-net cushion. |
| Integration and service disruption (UP/SP precedent) | Medium | Earnings and STB oversight risk | Service Assurance Plan, Safety Integration Plan, 5+ years of oversight, staged cut-over. |
| Leverage and rating pressure (~3.1–3.4x) | Low-medium | Higher funding cost; buybacks delayed | No financing condition; pre-close cash build; buybacks paused; “A” commitment. |
| Borrow cost or squeeze on the UNP short | Low | Hedge slippage | UNP is a liquid large-cap, and the 1.0 ratio is fixed, so the hedge stays exact to closing. |

Source: Merger agreement s.6.3, 7.1, 7.3; 49 CFR 1180.1; STB decisions in FD 36873; model sheets Arb, Scenarios, Synergies, Financing; Fitch rating watch (29-Jul-2025, 15-Jul-2026).

# Recommendation: support the deal as a UNP holder; go long NSC / short UNP at ~9% of NAV notional

Two separate views. The merger creates modest value for UNP holders on management's net synergies, and the spread prices too little chance of approval.

## Deal merit for UNP holders: qualified support

- Value created: +$6.9B (+$11.56/sh, +4.4%) at $2.0B net synergies; synergies cover the premium 1.53x.
- EPS: −1.2% / +5.2% / +9.4% in years 1–3, consistent with guidance.
- But the price is full (17.7x LTM EBITDA today) and cost synergies alone do not cover it. Break-even is ~$1.1B of net synergies.
- Watch: the conditions the STB attaches, evidence for revenue synergies in the record, and the leverage path to ~2.8x.

## Arb: long NSC / short 1.0 UNP

- Market-implied P(close) 25% vs our 55%; breakeven 25%.
- Payoff +$51.24 / −$17.08 per NSC share (3.0:1); expected +6.3% of notional over ~1 year.
- Size: 9.5% of NAV long notional, so a break costs 0.5% of NAV and a close adds ~1.5%.
- Add on a hearing order or DOJ/USDOT comments (3-Dec-2026) without broad remedies; cut on broad trackage-rights requests or a CSX/BNSF counter-deal.

**$11.56**
Value created per UNP share

**+9.4%**
Year-3 EPS accretion

**25% vs 55%**
Implied vs our P(close)

**+6.3%**
Hedged expected P&L, % notional

### WHAT WOULD CHANGE OUR MIND

Deal: STB conditions that cut net synergies below ~$1.1B, or a downgrade out of the A category. Arb: a schedule that pushes the decision into 2028, a UNP statement that conditions are unacceptable, or NSC above ~$345 with no news (implied P above ~50%).

Source: Model (all figures). The rules are the case study's own and are illustrative. Not investment advice.

# Appendix: key assumptions and items that could not be verified

Every input is in the Inputs sheet and figures.json with unit, as-of date and source URL, or is flagged as an assumption (28 assumptions).

## KEY ASSUMPTIONS

| Assumption | Value |
|---|---|
| Close: base / delayed / denial date | 31-Oct-27 / 30-Apr-28 / 30-Sep-27 |
| P(A) / P(B) / P(C) / P(D) | 47% / 8% / 5% / 40% |
| WACC inputs: beta / ERP / Kd | 1.0 / 4.25% / 5.9% |
| Exit multiple; NSC plan haircut | 14.5x; 6% |
| Synergy phase-in; growth; discount rate | 33/67/100%; 2.5%; 9% |
| New-debt coupon; pre-close cash build | 5.75%; $5B |
| UNP move on a break; loss budget | +3%; 0.5% of NAV |
| NSC discount to peer multiple | 6.2% |

## NOT VERIFIED FROM PRIMARY SOURCES, OR INTERPRETIVE

- NSC credit ratings and outlooks (Moody's Baa1 review): secondary source.
- CN multiples use FY2025 (no 2026 XBRL). BNSF EBITDA = operating income + D&A from BNSF LLC filings.
- NSC LTM adjusted EBITDA adds back company non-GAAP items (merger costs, restructuring, Eastern Ohio net recoveries).
- Close date and scenario probabilities are our judgement; the STB hearing date is not set.
- End Date extension mechanics and MBRC scope are read from the agreement; how they apply in practice could differ.
- Size of eventual STB concessions, and whether UP management's $2.0B net case (prepared mid-2025) is still current.
- CPKC application and voting-trust dates come from the STB decision summary, not a full re-read.

Source: figures.json; model/UNP_NSC_merger_model.xlsx (check.py: LibreOffice recalculation, 0 mismatches). Primary sources: SEC EDGAR (merger agreement, 424B3, 8-K, 10-K, 10-Q), STB Docket FD 36873, Federal Register, eCFR. Prices: Yahoo Finance.